

Feature-Based and Amino-acid based ClonalTree Report(Ongoing Research)

Project Status

Important Notice

This repository documents ongoing research on alternative approaches for B-cell lineage reconstruction using the ClonalTree framework.

The project is currently under active development and has not yet been completed. Some implementation details and experimental results are intentionally omitted because the work is ongoing and is being prepared for future publication.

The purpose of this report is to describe the motivation, methodology, and current progress of the project.

1 Introduction

B-cell lineage reconstruction is an important problem in computational immunology. During affinity maturation, B-cells accumulate mutations that generate diverse populations of related antibody sequences. Reconstructing the evolutionary relationships among these sequences provides insight into immune responses, clonal evolution, and antibody development.

Many existing lineage reconstruction methods rely primarily on nucleotide-level sequence similarity. ClonalTree is one such method that combines sequence distances, abundance information, and Minimum Spanning Tree (MST) construction to efficiently infer lineage relationships.

While nucleotide-based approaches are computationally efficient, they may not always capture biologically meaningful relationships between sequences. Mutations that appear significant at the nucleotide level may have little or no effect at the protein level, while biologically important amino-acid substitutions may not be adequately represented by simple sequence distance measures.

The objective of this research is to investigate whether alternative sequence representations can improve lineage reconstruction while maintaining the computational efficiency of ClonalTree. In particular, we explore three complementary approaches:

1. Traditional nucleotide-based lineage reconstruction. “
2. Amino-acid based lineage reconstruction using protein-level sequence similarity.
3. Biologically informed reconstruction methods that incorporate biological, chemical, and physicochemical properties of amino acids. “

The long-term goal is to evaluate how these different representations affect lineage tree topology, biological interpretability, and reconstruction quality.

2 Amino-Acid Based Reconstruction

2.1 Motivation

Traditional ClonalTree reconstruction relies on nucleotide-level Hamming distance. Although computationally simple and effective, nucleotide similarity does not always reflect functional similarity between proteins.

To investigate whether protein-level information can provide a more biologically meaningful representation of clonal evolution, we developed an amino-acid based version of the ClonalTree reconstruction pipeline.

2.2 Method

The LLC benchmark datasets provide aligned nucleotide sequences but do not directly provide amino-acid datasets suitable for ClonalTree analysis.

For this study, amino-acid sequences were obtained using the IMGT framework and used as input for amino-acid based lineage reconstruction.

The workflow consists of:

1. Obtaining amino-acid sequences from IMGT.

““

2. Computing pairwise amino-acid Hamming distances:

$$d(i, j) = \text{HammingDistance}(AA_i, AA_j)$$

3. Constructing an amino-acid distance matrix.

4. Applying the standard ClonalTree MST procedure:

$$\text{primMST}(AA_adjMatrix, root, labels, abundance)$$

5. Applying trimming and revision procedures when required.

6. Exporting the resulting lineage tree and branch-distance information. ““

This approach enables direct comparison between nucleotide-based and amino-acid based reconstruction strategies while preserving the underlying ClonalTree framework.

2.3 Example Output

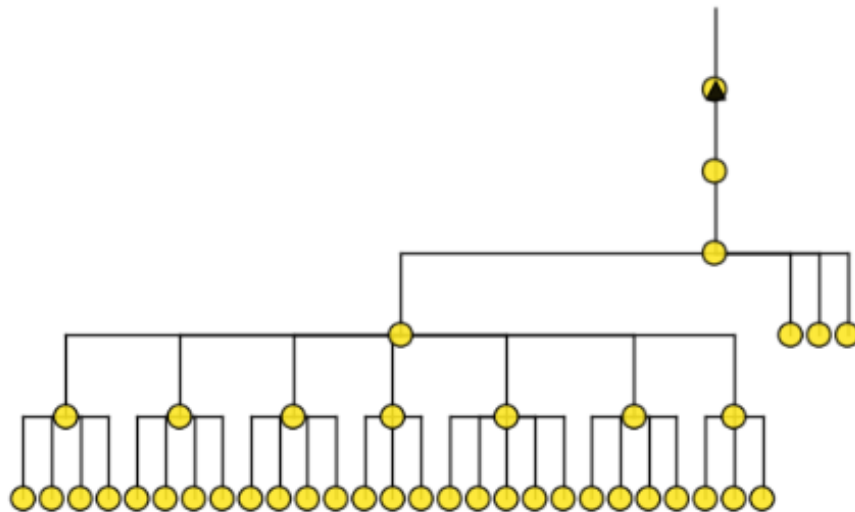


Figure 1: Amino-acid based lineage tree generated using ClonalTree and amino-acid Hamming distance.

3 Feature-Based Reconstruction

3.1 Initial Implementation

The first feature-based implementation was designed to replace sequence-based distances with distances computed from externally extracted numerical features.

Instead of representing each sequence by nucleotides or amino acids, each sequence was represented by one or more numerical descriptors obtained from a feature table. The resulting distances were then supplied to the standard ClonalTree MST construction procedure.

The initial implementation selected a single feature and computed pairwise distances using:

$$[d(i,j) = |f(i) - f(j)|]$$

where f denotes the selected feature.

Although this approach successfully generated lineage trees, it introduced an important limitation. Since all sequences were effectively projected onto a single dimension, MST construction frequently produced chain-like or stick-like topologies.

3.2 Initial Output

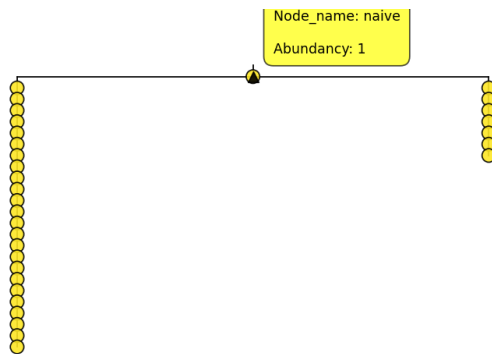


Figure 2: Initial feature-based reconstruction on Dataset 02 using a single-feature distance metric.

3.3 Improved Feature-Based Reconstruction

To address the limitations of the initial implementation, the distance construction procedure was redesigned to incorporate multiple features simultaneously.

Each sequence is represented by a multidimensional feature vector:

$$[X_i = (f_1(i), f_2(i), \dots, f_k(i))]$$

Distances are computed in feature space using:

$$[d(i,j) = \sqrt{\sum_{m=1}^k (X_{i,m} - X_{j,m})^2}]$$

Several additional improvements were introduced:

- Robust matching between FASTA sequence identifiers and feature-table identifiers. ““
- Improved handling of missing feature values.
- Consistent abundance labeling after tree construction.
- More stable MST generation using multidimensional feature representations. ““

These modifications significantly reduced the tendency of the MST to collapse into chain-like structures and produced more informative lineage topologies.

3.4 Improved Outputs

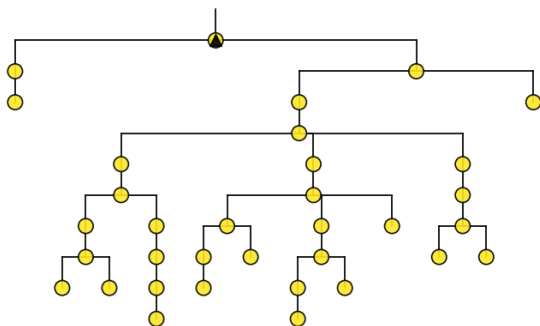


Figure 3: Feature-based reconstruction on Dataset 01 using multidimensional feature vectors and Euclidean distance.

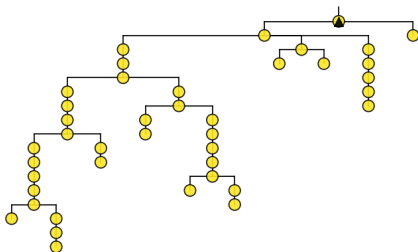


Figure 4: Feature-based reconstruction on Dataset 02 using multidimensional feature vectors and Euclidean distance.

4 Biologically Informed Reconstruction (Ongoing Work)

While amino-acid Hamming distance captures protein-level differences, it still treats all amino-acid substitutions equally.

The current stage of the project focuses on developing biologically informed similarity measures that incorporate biological, chemical, and physicochemical properties of amino acids.

The motivation behind this approach is that different amino-acid substitutions can have substantially different biological consequences. Some substitutions preserve important structural and functional characteristics, whereas others may significantly alter protein behavior.

The goal is therefore to develop efficient lineage reconstruction methods that incorporate biologically meaningful information while preserving the computational advantages of the ClonalTree framework.

Because this component is still under active development, detailed methodology and experimental results are not included in this report.

5 Current Status

At the current stage, the project includes:

- Nucleotide-based ClonalTree reconstruction.
- Amino-acid based ClonalTree reconstruction.
- Feature-based ClonalTree reconstruction.
- Development of biologically informed similarity measures.
- Benchmark evaluation on LLC datasets.

6 Future Work

Future work will focus on comprehensive comparison of reconstruction strategies, evaluation on benchmark datasets, further development of biologically informed similarity metrics, and preparation of results for publication.